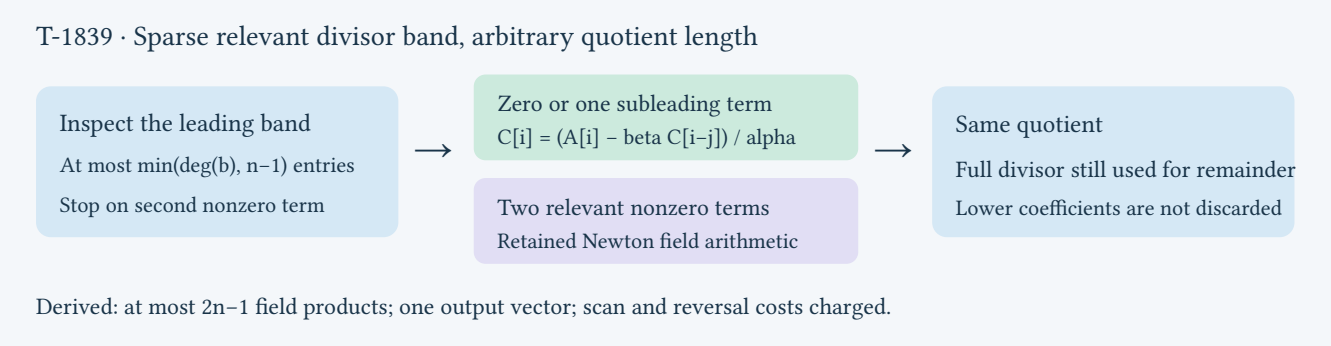
Linear recurrence for sparse quotient bands

T-1839 · 2026-11-07 · Arbitrary quotient length

Quotient construction now recognizes a sparse relevant leading band of the divisor. For quotient length n, only the top n divisor coefficients affect the quotient. With at most one nonzero subleading coefficient in that band, a delayed recurrence replaces inverse-series construction for any n. Lower divisor coefficients still participate in the remainder.



Identity. Write the reversed divisor modulo X^n as alpha + beta X^j, with alpha nonzero. The reversed quotient satisfies $C_i=(A_i-\text{beta } C(i-j))/\text{alpha}$, using zero for negative indices. Induction gives B C=A modulo X^n and therefore the same quotient as Newton. An absent beta gives independent coefficient scaling.

Exact cost. Let u indicate alpha≠1. Skipping zero products gives $M=u\,Z(C)+Z_j(C)$, where Z counts nonzero quotient coefficients and Z_j counts those in the first n−j reversed positions. Thus $M\le u\,n+n-j\le 2n-1$, with u inversions. Without a subleading term, $M\le u\,n$. Scan costs at most min(deg(b),n−1) comparisons; output reversal is linear. The routine keeps one n-entry output buffer and constant scan state, while input/remainder storage remains separate.

Fallback. A second relevant nonzero coefficient stops inspection and uses the prior Newton arithmetic. Its added scan cost is explicitly charged. Length-one/two formulas and zero dividends retain their prior paths. No universal timing advantage or source-workload frequency is asserted.

Exact software check	Count	Coverage
Sparse leading-band cases	1,536	Outputs; exact products; scan bounds
Lower / equal / higher products	1,536 / 0 / 0	Fixed quotient-stage panel
Dense fallback controls	384	Identical field counts; two inspections
Lower-coefficient controls	1,536	Same quotient; changed remainder
Source points / windows	1,108 / 24	Direct products, covariance, signs
Additional recoveries	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 calls

Evidence and scope. Retained short/wider quotient, division, matrix, resultant, singleton and tree controls pass. Normal build pins seven native inputs without audits. Original 4,563-case cohort unchanged. No cloud, timing benchmark, external comparison or full Cargo rerun. PCC-20261107-44be6f is unsigned producer evidence. DERIVATION.md and receipts contain assumptions, commands and hashes.

Sources and next work. Standard sparse series recurrence and FLINT binomial-inverse documentation, with exact predecessor code; KB unavailable. No novelty claim. Next trace support sizes and quotient lengths in the actual source/recovery path before choosing a larger support cap. Historical reports and canonical graphs remain unchanged.